Cost-Latency Tradeoff in LLM-Based Systems: Comparing API-Based and Local Models under Prompting, RAG, and Context Optimization

Buddhima Sandaru Senarathna
Department of Computer Science
University of Helsinki
Helsinki, Finland
buddhima.senarathna@helsinki.fi

***Abstract*—Large Language Models (LLMs) face significant cost and latency challenges in production deployments, particularly for long regulatory documents. This paper compares three architectures–prompt-only, Retrieval-Augmented Generation (RAG), and context-optimized systems–across cloud API (Gemini 2.5 Flash) and local open-weight (Qwen 2.5 1.5B) deployments using European Union (EU) regulatory question and answering (QA) tasks.**

Results show prompt-only systems are infeasible for local deployment (0% Exact Match (EM), 67 s latency) and costly for API use (79k tokens/query). RAG achieves the lowest latency (3.3 s) and tokens (785–2.7k) across both but sacrifices EM accuracy. Context-optimized systems provide the best trade-off for API deployment, matching prompt-only EM (80%) with 40 % fewer tokens.

The findings demonstrate no universal optimum: RAG excels in resource-constrained local settings while context optimization balances accuracy and cost for API deployments. These results provide practical engineering guidelines for LLM system design beyond accuracy alone, emphasizing cost, latency, and hardware feasibility as first-class objectives.

***Index Terms*—Retrieval-Augmented Generation, context optimization, LLM deployment, cost-latency tradeoff, regulatory QA**

I. Introduction

Large Language Models (LLMs) have become central to natural language processing systems, enabling document-based question answering, summarization, and conversational agents. While recent advances focus on improving accuracy and reasoning, practical deployment introduces challenges related to *computational cost*, *latency*, and *resource utilization*. In production settings, inference costs scale with token usage and response latency directly impacts user experience [1], [2]. Modern deployments increasingly rely on both cloud-based API models and locally hosted open-weight models, each with distinct cost, scalability, and latency trade-offs.

Retrieval-Augmented Generation (RAG) improves LLM performance on knowledge-intensive tasks by retrieving relevant documents from external knowledge bases [3]. RAG enhances factual accuracy and reduces hallucinations compared to prompt-only approaches [4], but introduces embedding models, vector databases, and retrieval pipelines that add computational overhead [5].

Despite extensive RAG research, *there is limited empirical analysis comparing prompt-only, RAG-based, and simpler context-optimized approaches from a cost-latency perspective*. Prompt-only systems embed all information directly into prompts (minimal overhead but high token usage for long documents), while RAG trades complexity for grounding, and emerging long-context LLMs question retrieval necessity [7], [8].

This paper investigates *how LLM system designs trade off cost, latency, and quality under realistic constraints*, comparing: (1) prompt-only, (2) RAG-based, and (3) context-optimized systems. We evaluate token usage, end-to-end latency, and output quality on regulatory QA tasks across API and local deployments.

Our findings show RAG improves quality in some settings but context-optimized systems often achieve competitive performance with lower cost and latency, providing practical guidelines for efficient LLM system design beyond accuracy alone.

II. System Design and Experimental Setup

This chapter presents the system architectures, language models, corpus construction, and evaluation protocol used in this study. The objective is to compare alternative long-context processing strategies for regulatory question answering under realistic deployment constraints.

A. Large Language Models

To reflect realistic deployment scenarios, this study evaluates two fundamentally different categories of LLMs:

- A **cloud-based API model**
- A **locally deployed open-weight model**

This distinction enables analysis not only of architectural design choices (prompt-only, RAG, context optimization), but also of deployment trade-offs between external inference services and on-device computation.

1) Gemini 2.5 Flash (API-based Model): The first model used was `Gemini 2.5 Flash`, accessed via a cloud-based

API. This model represents a highly optimized, production-grade LLM designed for low-latency inference and efficient token usage.

As an API-based model, it abstracts away hardware constraints and benefits from large-scale infrastructure optimizations. However, it introduces *monetary cost per token* and *network latency*, which must be considered in real-world deployments.

2) Qwen 2.5 1.5B Instruct: The second model was `Qwen/Qwen2.5-1.5B-Instruct`, a 1.5B-parameter instruction-tuned decoder-only transformer with a 32,000-token context window.

This model was deployed locally on GPU hardware, enabling full control over inference without per-token API costs. However, performance is constrained by available computational resources, particularly GPU memory and processing speed.

3) Comparison Rationale: The inclusion of both models enables a realistic comparison between:

- **API-based inference**: scalable, optimized, but cost-incurring
- **Local inference**: cost-efficient at scale, but hardware-limited

All experiments were conducted with temperature fixed at 0 to ensure deterministic outputs and fair comparison.

B. System Architectures

Three representative LLM-based architectures were designed and evaluated:

- Prompt-Only
- Retrieval-Augmented Generation (RAG)
- Context-Optimized

Each system architecture was evaluated across both models. The architectures differ in how contextual information is constructed and supplied, while the underlying model type (API vs local) introduces additional differences in latency and cost characteristics.

1) Prompt-Only System: The prompt-only system embeds the entire regulatory document directly into the model prompt together with task instructions and the user query (Fig. 1). The model is explicitly instructed to extract the exact answer span from the provided text.

This architecture consists of a single inference step without preprocessing or external retrieval components [1], [2]. However, token usage grows linearly with document length and is bounded by the model's maximum context window [8]. No context filtering or compression is applied.

2) Retrieval-Augmented Generation (RAG): The RAG system follows a two-stage retrieval–generation pipeline (Fig. 2) [3], [4].

a) Document Chunking: Each regulatory document was segmented into fixed-length chunks of 256 tokens with overlapping windows to mitigate boundary information loss. Overlapping chunking is standard practice in retrieval-based QA systems [3], [15]. The chunk size was selected to preserve article-level semantic coherence while remaining compatible with GPU memory constraints.

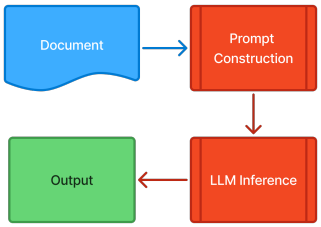

Fig. 1. Prompt-only system architecture.

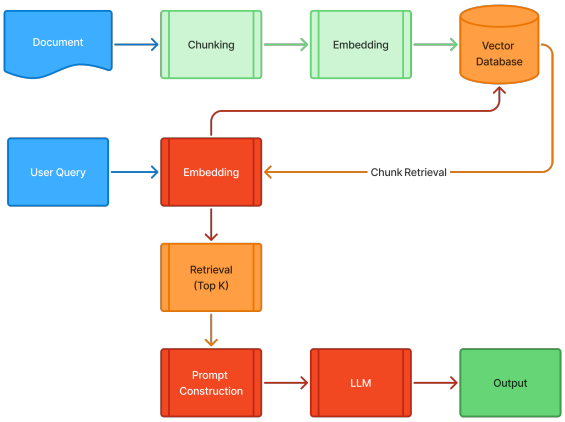

Fig. 2. Retrieval-augmented generation (RAG) system architecture.

b) Embedding Model: Chunks were embedded using the `sentence-transformers/all-MiniLM-L6-v2` model, a lightweight 384-dimensional dense embedding model. This model was selected due to its favorable trade-off between semantic similarity quality and computational efficiency.

c) Vector Indexing and Retrieval: Embeddings were indexed using FAISS with L2 similarity search. At inference time, the user query was embedded and used to retrieve the top-$k = 3$ most similar chunks. The retrieved chunks were concatenated with the task prompt and passed to the LLM for answer extraction.

While RAG reduces irrelevant context exposure, it introduces additional components including embedding computation and similarity search, which contribute to latency and system complexity [5], [6].

3) Context-Optimized System: The context-optimized system is designed as a lightweight alternative positioned between prompt-only and RAG-based approaches (Fig. 3).

Instead of performing external retrieval, the full document

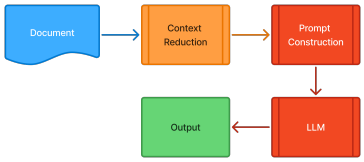

Fig. 3. Context-optimized system architecture.

is first segmented into chunks and scored using embedding similarity with respect to the query. All chunks are ranked globally, and only the highest-scoring segments within a predefined token budget are selected and concatenated into a structured context block supplied to the LLM.

Unlike RAG, this approach does not retrieve a small fixed number of local chunks. Instead, it performs global relevance ranking and selective context construction, preserving broader document coherence while filtering irrelevant sections.

a) Design Rationale: EU regulatory texts such as GDPR, the AI Act, the DSA, the DMA, and the Data Governance Act are long, hierarchically structured, and dense in cross-references. Naively inserting full documents introduces noise and token inefficiency, while standard RAG may fragment logically connected provisions.

Recent work shows that long-context LLMs do not utilize all tokens uniformly and may suffer from attention degradation in the middle of long inputs [8]. The context-optimized design therefore aims to balance:

- Token efficiency,
- Semantic relevance,
- Document coherence,
- Reduced hallucination risk.

C. Regulatory Document Corpus

The evaluation corpus consists of five long-form European Union regulatory documents:

- General Data Protection Regulation (GDPR) [9]
- Artificial Intelligence Act [10]
- Digital Services Act (DSA) [11]
- Digital Markets Act (DMA) [12]
- Data Governance Act [13]

These documents were selected due to their length, hierarchical structure, dense legal language, and cross-referencing, which present significant long-context reasoning challenges.

Each document was converted into structured plain text while preserving article numbering and paragraph hierarchy.

A dataset of 100 manually annotated extractive question-answer pairs (approximately 20 per document) was constructed. Questions cover definitions, obligations, prohibitions, numerical thresholds, and cross-references. Ground-truth answers were annotated as exact spans within the source documents.

D. Evaluation Metrics

The primary objective of this study is efficiency analysis. Therefore, token cost and latency are treated as primary metrics.

Token Usage: For each query, total token consumption is defined as

$$Tokens_{total} = Tokens_{input} + Tokens_{output} \tag{1}$$

Since transformer inference complexity scales quadratically with input sequence length due to self-attention mechanisms [16], prompt length directly impacts computational cost.

Latency: End-to-end latency is measured using wall-clock timing:

$$T_{total} = T_{preprocess} + T_{retrieval} + T_{generation} \tag{2}$$

Latency measurement includes preprocessing, embedding computation (for RAG), vector retrieval, and language model generation. End-to-end measurement is standard in retrieval system evaluation [15].

Exact Match (EM): Since the task is extractive, answer correctness is evaluated using Exact Match following standard QA manual evaluation methodology [**?**].

E. Hardware Configuration

The local LLM (Qwen 2.5 Instruct) was executed on a system equipped with an NVIDIA RTX 4050 laptop GPU (6 GB VRAM), an Intel i7-13620H CPU, and 16 GB RAM.

The API-based model (Gemini 2.5 Flash) was accessed via a cloud service and therefore did not rely on local hardware for inference. However, end-to-end latency measurements include network communication overhead in addition to model response time.

All latency measurements were performed under stable network conditions to ensure consistency.

F. Experimental Hypotheses

The following hypotheses are evaluated:

- H1: API-based models achieve lower latency due to infrastructure optimization but incur monetary cost per query.
- H2: Local models exhibit higher latency but eliminate per-query cost.
- H3: RAG reduces token usage but introduces additional latency regardless of model type.
- H4: Context-optimized processing provides the best cost–latency tradeoff across both deployment settings.

III. Experimental Results

This chapter presents the empirical evaluation of the three system architectures across the regulatory corpus for both the local Qwen model and the API-based Gemini model. The analysis focuses on token usage, latency, and answer correctness, and examines the cost-latency trade-offs implied by these measurements.

TABLE I
QWEN: SUMMARY PERFORMANCE ACROSS ALL REGULATIONS

System	Avg. EM (%)	Avg. Latency (s)	Avg. Tokens
Context-Filtered	59.0	4.69	2521.95
Prompt-Only	0.0	67.02	4096.00
RAG	73.0	3.32	785.17

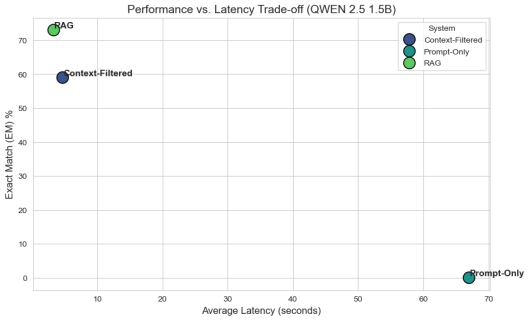

Fig. 4. Qwen: EM vs. latency across system architectures.

TABLE II
GEMINI: SUMMARY PERFORMANCE ACROSS ALL REGULATIONS

System	Avg. EM (%)	Avg. Latency (s)	Avg. Tokens
Context-Filtered	80.0	4.87	45264.56
Prompt-Only	80.0	5.68	79040.11
RAG	67.0	3.02	2665.14

A. Overall Performance Comparison

1) Qwen (Local Model): Table I summarizes the performance of the three systems when using the locally hosted Qwen 2.5 1.5B Instruct model.

The prompt-only system failed to produce any exact matches under the Qwen setup, primarily due to hard token limits and hardware constraints which prevented full document prompts from being processed. In practice, many queries approached or exceeded the maximum context window and output token limit, causing truncation and incomplete reasoning. By contrast, both RAG and context-filtered systems substantially reduced input length and achieved non-zero Exact Match scores, with RAG reaching the highest EM (73%) and the lowest average latency (3.32 s).

The context-filtered system achieved moderate EM (59%) with considerably lower latency than prompt-only and substantially fewer tokens than a full-document prompt. Qualitatively, RAG answers often captured the correct *idea* of the target span (e.g., paraphrased obligations or conditions) even when they missed the exact wording required for EM, while the context-filtered system more frequently reproduced the precise extractive span due to the inclusion of larger coherent sections of the originating articles. A summary plot of EM versus latency for Qwen (Figure 4) further illustrates the dominance of RAG and context-filtered approaches over prompt-only in the local deployment regime.

2) Gemini (API-Based Model): Table II reports the corresponding results for the Gemini 2.5 Flash API-based model.

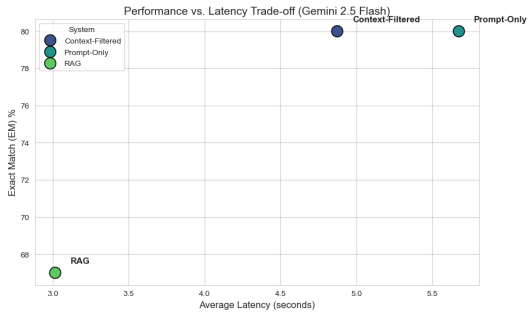

Fig. 5. Gemini: EM vs. latency across system architectures.

For Gemini, both context-filtered and prompt-only systems achieved the same average EM of 80%, indicating that the API-based model can handle very long prompts more robustly within its larger context and output limits. However, this accuracy parity hides substantial differences in efficiency: prompt-only required on average almost 80k tokens per query, while context-filtered reduced token usage by more than 40% and slightly improved latency.

The RAG system, again, achieved the lowest token usage and the fastest average latency (3.02 s) but at the cost of lower EM (67%). As with Qwen, RAG answers were often semantically close to the ground-truth span yet failed the strict extractive EM criterion. Figure 5 shows the EM-latency trade-off for Gemini, highlighting the different accuracy-efficiency regimes occupied by the three architectures.

B. Prompt-Only Analysis

For the Qwen deployment, prompt-only performance was dominated by context window and compute constraints. Long EU regulations easily approach or exceed several thousand tokens when entire articles or recitals are included, and the 32k-token context window combined with a relatively small GPU (6 GB VRAM) forced aggressive truncation of inputs and outputs. As a result, the model often saw only partial documents and failed to include the answer span within the visible context. This explains the observed 0% EM despite very high average latency (67.02 s), as Qwen had to process near-maximum-length inputs under limited hardware.

By contrast, Gemini's managed infrastructure allowed prompt-only inputs to include much larger portions of the regulatory texts without hitting hard context or output limits, leading to substantially higher EM (80%) at a still acceptable latency of 5.68 s on average. Nevertheless, the token counts (around 79k tokens per query) imply significant monetary cost and raise concerns about scalability for production deployments where millions of queries may be processed monthly.

Across documents (GDPR, AI Act, DSA, DMA, DGA), the prompt-only system tended to perform better on shorter or structurally simpler texts (e.g., parts of the DGA) and worse on long, highly cross-referenced regulations (e.g., the AI Act), where the relevant provisions could not always be included in a single contiguous prompt window.

C. Retrieval Performance (RAG)

For both models, RAG substantially reduced average token usage by restricting the context to the top-3 retrieved chunks. This reduction was most pronounced for Gemini, where the average token count dropped from tens of thousands (prompt-only/context-filtered) to approximately 2,665 tokens per query, with a concurrent reduction in latency to around 3 s.

Retrieval Recall@5 was generally high for questions whose answers were localized within a single article or a small cluster of adjacent paragraphs. In these cases, the correct chunk was often retrieved, and the model produced answers that were semantically faithful to the ground-truth span. However, RAG struggled with questions requiring dispersed evidence (e.g., definitions located in one article and conditions in another) or cross-references spanning multiple titles or chapters, which is common in EU legislation [9], [11]–[13]. In such cases, Recall@5 dropped, and the answer either missed critical conditions or hallucinated unsupported constraints, consistent with prior observations about retrieval failures in knowledge-intensive tasks [3], [15].

Even when retrieval succeeded, EM sometimes remained 0 because the generator produced paraphrased explanations rather than verbatim spans. Since EM is a strict extractive metric [**?**], these paraphrases were evaluated as incorrect despite being practically useful from a legal-interpretability perspective.

D. Context-Optimized System Analysis

The context-optimized system lies between prompt-only and RAG in both complexity and information access. It scores all chunks globally, then builds a context within a token budget while preserving larger contiguous spans of the regulation to reduce attention dilution in long inputs [8].

With Qwen, it reached 59% EM at 4.69,s and about 2,522 tokens, versus 0% EM and 67.02,s for prompt-only, but still below RAG (73% EM, 785 tokens, 3.32,s). For Gemini, it achieved the best EM (80%), matching prompt-only but with lower latency (4.87,s vs. 5.68,s) and far fewer tokens (about 45k vs. 79k), while outperforming RAG on EM (67%).

Qualitatively, it tends to include entire articles or related paragraphs, helping resolve cross-references and multi-clause conditions in EU regulations [9]–[13]. Overall, context optimization is particularly attractive in API settings where tokens are costly but long-context processing is available, offering a strong accuracy–efficiency trade-off and aligning with recent work on structured prompting for long legal documents [3], [7], [8].

E. Context Length Sensitivity

Across both deployment settings, latency scaled strongly with input token length, as expected from the quadratic complexity of self-attention with respect to sequence length [16]. For Qwen, this effect was amplified by GPU memory limits: near-maximal context lengths led to frequent paging and slower generation, which is reflected in the 67 s average latency for prompt-only versus 3–5 s for RAG and 4–7 s for context-filtered systems.

For Gemini, latency remained within a relatively narrow band (3–6 s) despite large variations in token counts, reflecting the benefits of optimized cloud infrastructure and possible internal batching and model parallelism. However, even in this regime, shorter prompts consistently yielded lower latency, and RAG's minimal token usage translated into the fastest responses.

Answer correctness showed a non-linear relationship with context length. Extremely short contexts (e.g., under-retrieved RAG chunks) lacked sufficient information and led to hallucinations or partial answers. Conversely, excessively long contexts (full documents in prompt-only) sometimes diluted attention around the relevant span, especially in the local model, consistent with prior work on long-context degradation [8]. The context-optimized system, which enforced a token budget while preserving semantically relevant spans, struck a balance, achieving high EM with moderate token counts, particularly for Gemini.

F. Cost-Latency Comparison

The measured token usage and latency can be used to reason about operational cost. For the API-based Gemini model, monetary cost scales approximately linearly with the number of input and output tokens, with per-token pricing set by the provider.[1] Under this model, the prompt-only system is the most expensive, consuming roughly 79k tokens per query. The context-filtered system reduces token usage to about 45k tokens, which corresponds to an approximate 40% cost reduction for similar EM. RAG further reduces tokens to roughly 2.7k per query, representing an order-of-magnitude cost reduction at the expense of lower EM.

In contrast, the local Qwen deployment has near-zero marginal cost per additional token once hardware is provisioned, but latency and feasibility are constrained by GPU memory and throughput. Here, the prompt-only system is not just slow but effectively unusable due to timeouts and context truncation, while RAG and context-filtered systems deliver acceptable latency (3–5 s) and non-zero EM without additional monetary cost. This aligns with broader observations that local deployments favor architectures that aggressively control context length in order to fit hardware budgets, whereas API-based deployments must balance per-token cost against the benefits of longer context [3], [7], [15].

From a cost-latency perspective, RAG occupies the "low-cost/low-latency but medium-accuracy" corner, while prompt-only sits at "high-cost/high-latency but potentially high-accuracy" (for sufficiently capable APIs), and context-filtered lies in a middle regime that offers competitive accuracy with moderate cost and latency across both deployment settings.

[1]Exact pricing is omitted here for generality; practitioners can plug in their own cost schedule.

IV. Discussion

A. Simplicity vs. Efficiency

Prompt-only is architecturally simple but either infeasible (local Qwen) or expensive (Gemini) for long regulations. RAG adds engineering overhead (embeddings, indexing, context scoring) and context optimization also adds context scoring but drastically improve latency and, in many cases, accuracy or feasibility [3]–[5].

B. Long Context vs. Selection

The results indicate that simply increasing context length is not sufficient. Long prompts can dilute attention and waste tokens, especially on dense legal texts [8]. RAG aggressively reduces context, trading some recall for speed and cost, whereas context optimization strikes a middle ground: fewer tokens than full documents, but more coherent spans than a small set of local chunks, which is beneficial for cross-referential texts [9], [11]–[13].

C. Deployment Considerations

For local deployments, RAG or context-filtered variants are effectively mandatory, as naive long-context prompting is too slow or fails. For API deployments, context optimization appears attractive when exact answers and traceability matter, while RAG is preferred for low-cost, low-latency scenarios where small drops in EM are acceptable.

V. Limitations

The study is limited by access to commercial APIs: free or low-cost tiers constrain the number of queries, context size, and model variants that can be tested. Local experiments are constrained by hardware; only a 1.5B model on a 6 GB GPU was evaluated, and larger open-weight models were excluded for practical reasons. Attempts to run larger Hugging Face models on CPU-only HPC infrastructure (such as cPouta) resulted in estimated runtimes on the order of thousands of hours, making such configurations infeasible for this kind of study. Finally, evaluation is restricted to extractive QA over EU regulations with EM; paraphrased but correct answers and other task types (e.g., summarization) were not systematically assessed but were manually corrected.

VI. Future Work

Future work includes evaluating larger local models and specialized long-context architectures, and combining RAG with context optimization under a joint token budget. More realistic cost models (including batching and multi-tenant workloads) and broader tasks such as summarization and scenario-based compliance analysis should be explored. Better human expert evaluation could complement EM to better capture the usefulness of near-exact or paraphrased answers in legal settings.

VII. Conclusion

This work empirically tested the four hypotheses outlined in Section II.G across Qwen (local) and Gemini (API) deployments on regulatory QA.

H1: "API-based models achieve lower latency due to infrastructure optimization but incur monetary cost per query."
Supported. Gemini consistently showed 3.6 s latency across all systems versus Qwen's 3.67 s range, but token usage (up to 79k/query) implies substantial per-token API costs.

H2: "Local models exhibit higher latency but eliminate per-query cost."
Supported. Qwen eliminated monetary cost but suffered high latency (67 s for prompt-only) due to GPU constraints, while RAG and context optimization reduced it to 3.5 s.

H3: "RAG reduces token usage but introduces additional latency regardless of model type."
Partially supported. RAG achieved the lowest tokens (785 for Qwen, 2.7k for Gemini) but also the lowest latency (3.32 s Qwen, 3.02 s Gemini), showing retrieval overhead was offset by shorter generation on optimized contexts.

H4: "Context-optimized processing provides the best cost–latency tradeoff across both deployment settings."
Partially supported. For Gemini, context optimization matched prompt-only EM (80%) with 40 % fewer tokens and slightly better latency. For Qwen, it improved from unusable prompt-only (0% EM) to 59% EM but was outperformed by RAG.

Overall, no single architecture universally dominates. RAG excels in latency-constrained local deployments, context optimization balances accuracy and cost in API settings, and naive prompting is rarely optimal for long regulatory texts. These results emphasize evaluating LLM systems on cost, latency, and hardware feasibility alongside accuracy when designing production regulatory QA pipelines.

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APPENDIX: CODE AND DATA AVAILABILITY

The code, experimental setup, datasets, and processing scripts used in this study are available in the author's public repository:

GitHub Repository:
https://tinyurl.com/AIENGSEM